



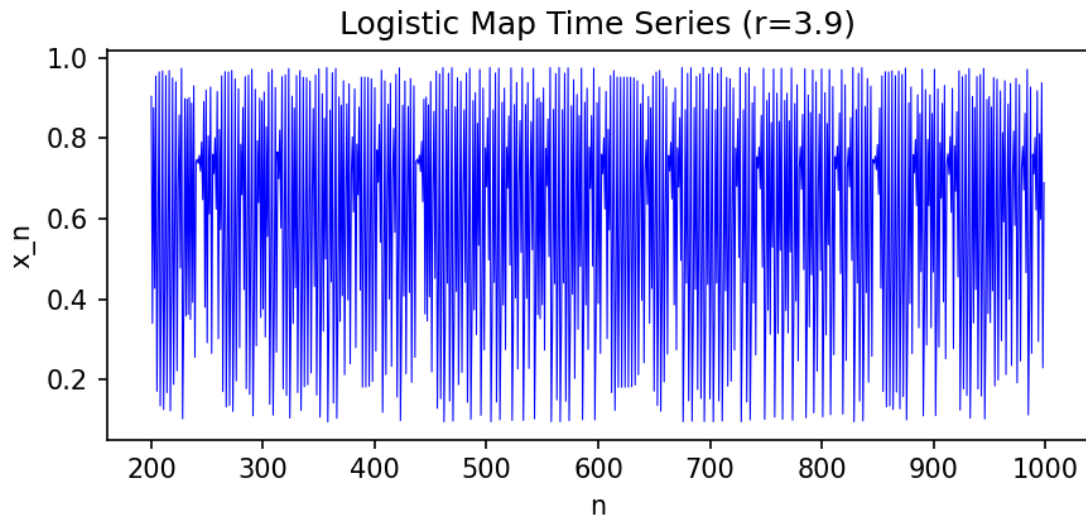
Task 1 report

Name: Kareem Yasser Mazrou

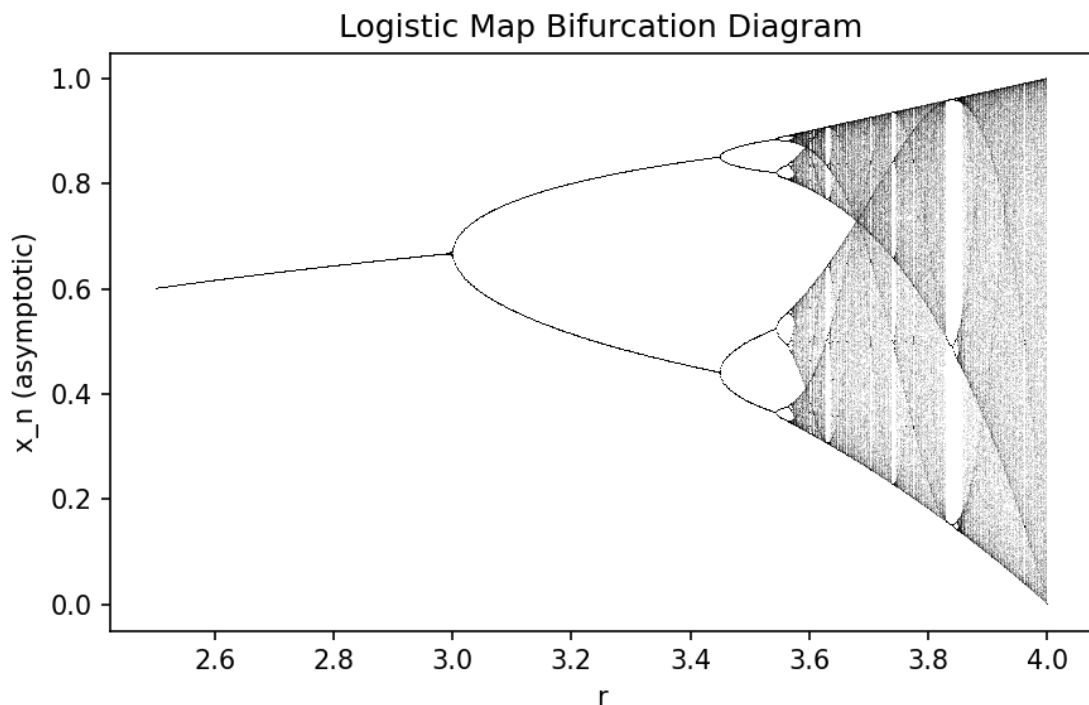
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Part A: Discrete Chaotic Systems

1. Logistic Map: The logistic map is a simple mathematical model of population growth that can show chaotic behavior. Below is the time series plot showing its behavior for a growth rate $r=3.9$.

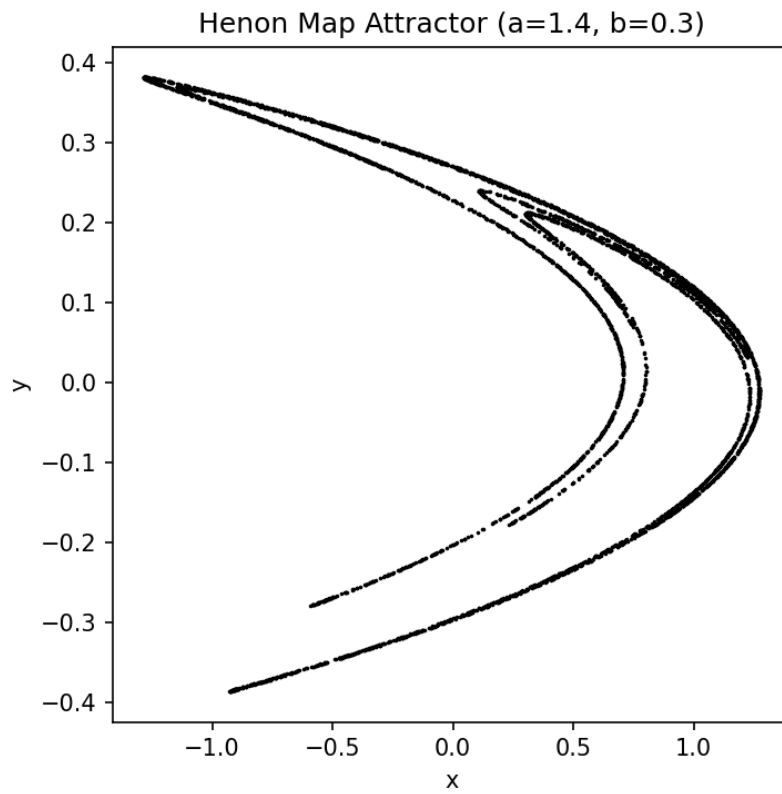


The bifurcation diagram below shows how the system's long-term behavior changes as we increase the growth rate r . It starts with a stable state, splits into periodic behavior (period doubling), and eventually becomes chaotic (the dense shaded areas) as r gets closer to 4.0.

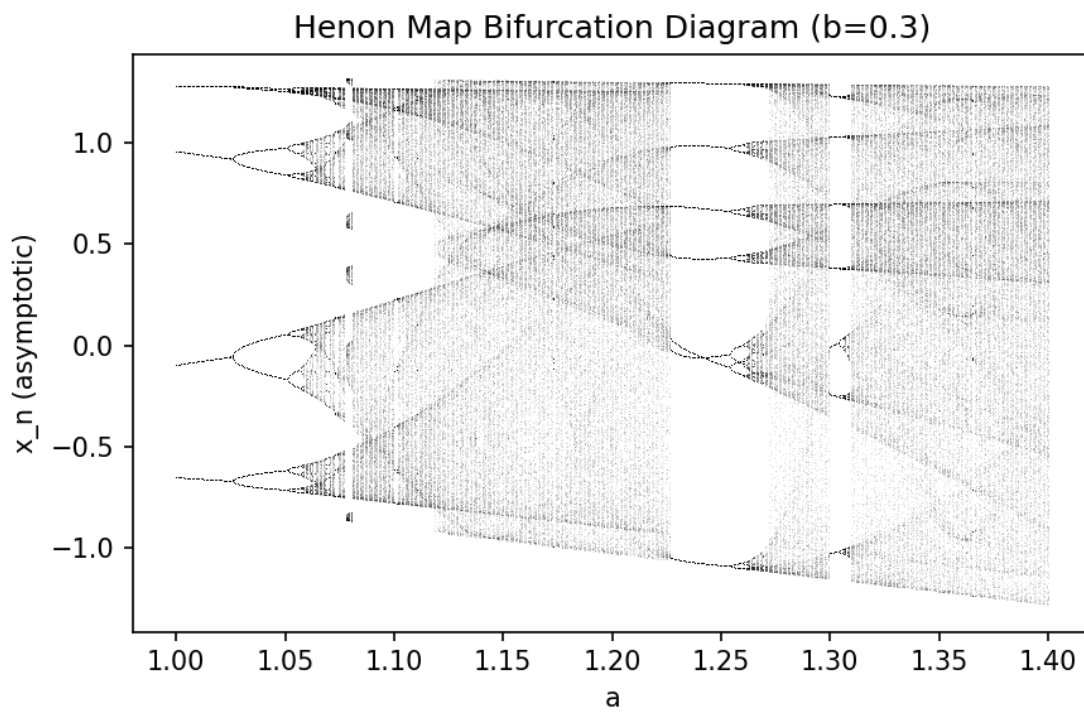


2. Henon Map: The Henon map is a 2D map that creates a classic fractal shape known as a strange attractor. The plot below shows this complex structure.

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The Henon map bifurcation diagram shows its transition to chaos as parameter 'a' increases. Similar to the logistic map, it undergoes period-doubling bifurcations before entering a chaotic state.



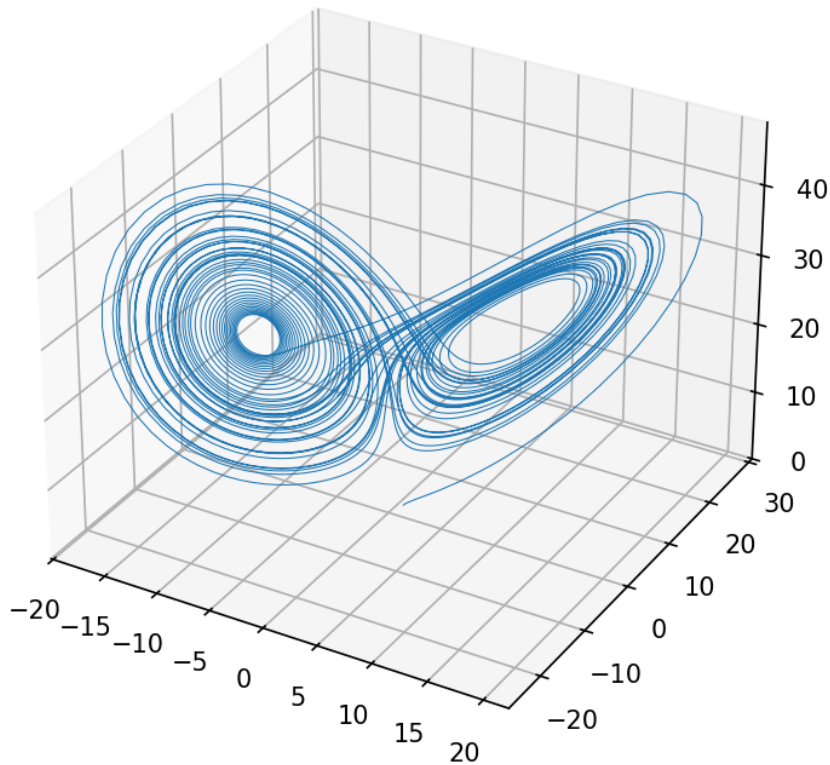
Part B: Continuous Chaotic Systems

1. Lorenz System

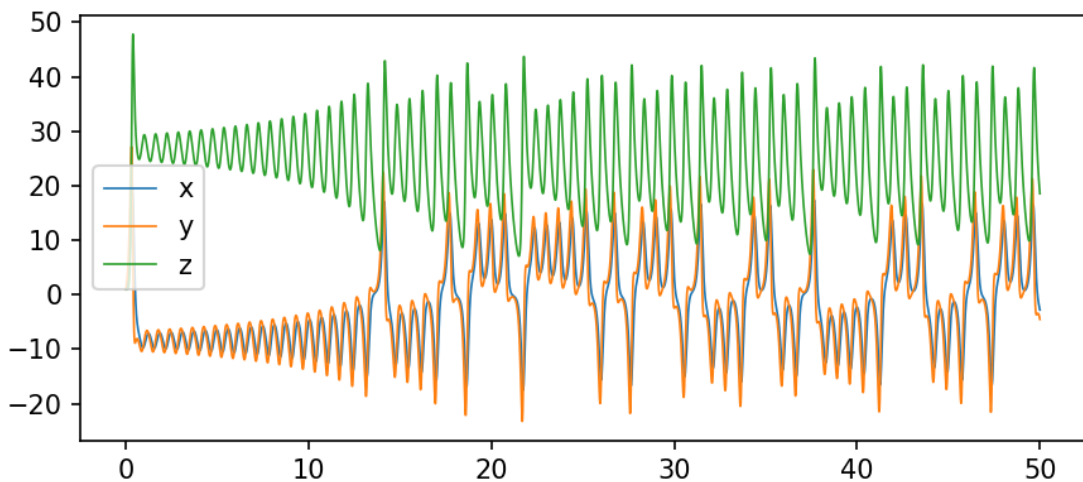
Equations: $dx/dt = \sigma(y - x)$, $dy/dt = x(\rho - z) - y$, $dz/dt = xy - \beta z$

Parameters: $\sigma=10$, $\rho=28$, $\beta=8/3$ | Initial Condition: (1,1,1) | $dt=0.01$ | Length: 50 | Solver: RK45

Lorenz Phase Space



Lorenz Time Series



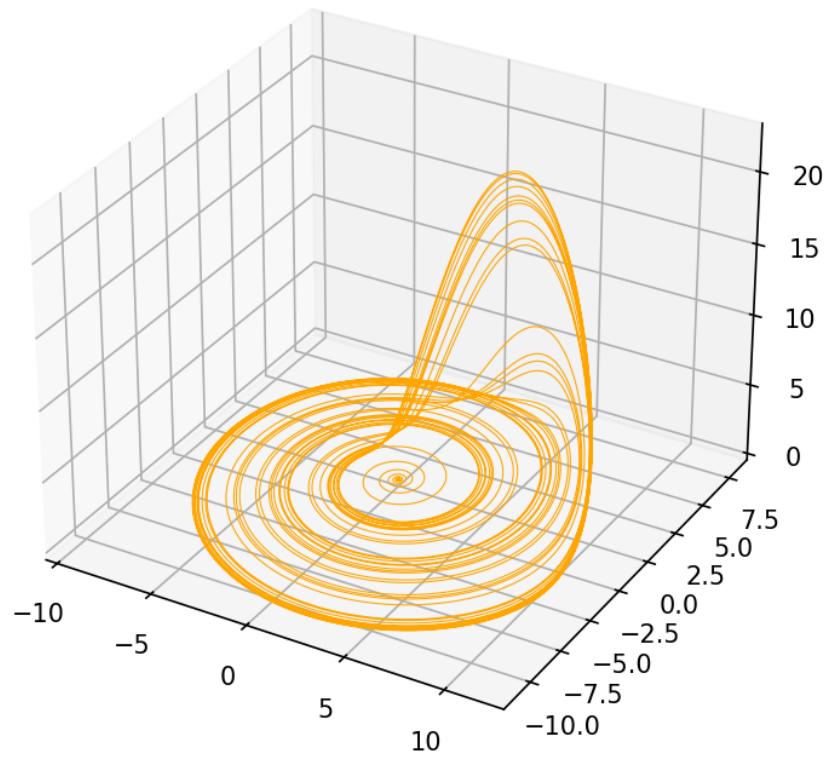
2. Rössler System

Equations: $dx/dt = -y - z$, $dy/dt = x + ay$, $dz/dt = b + z(x - c)$

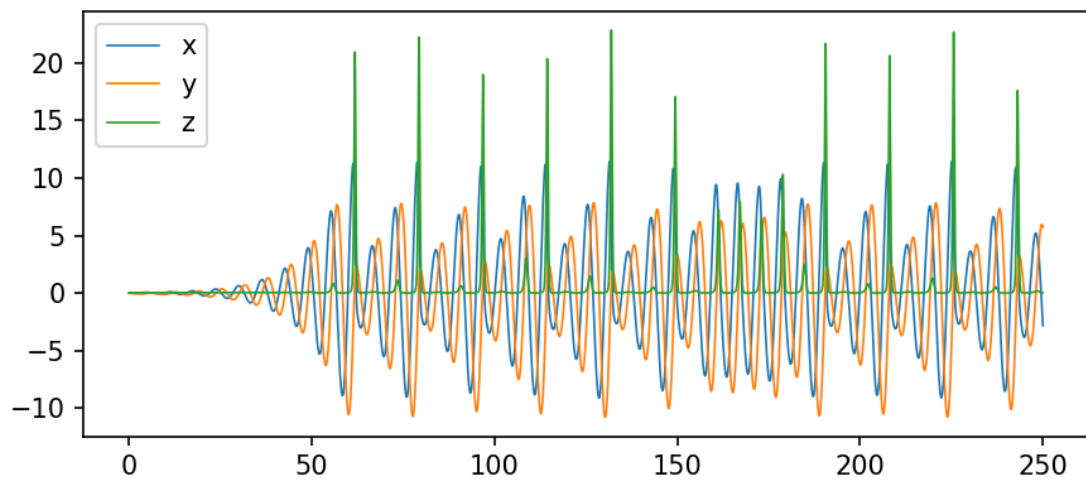
Parameters: $a=0.2$, $b=0.2$, $c=5.7$ | Initial Condition: (0,0,0) | $dt=0.01$ | Length: 250 | Solver: RK45

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Rössler Phase Space



Rössler Time Series



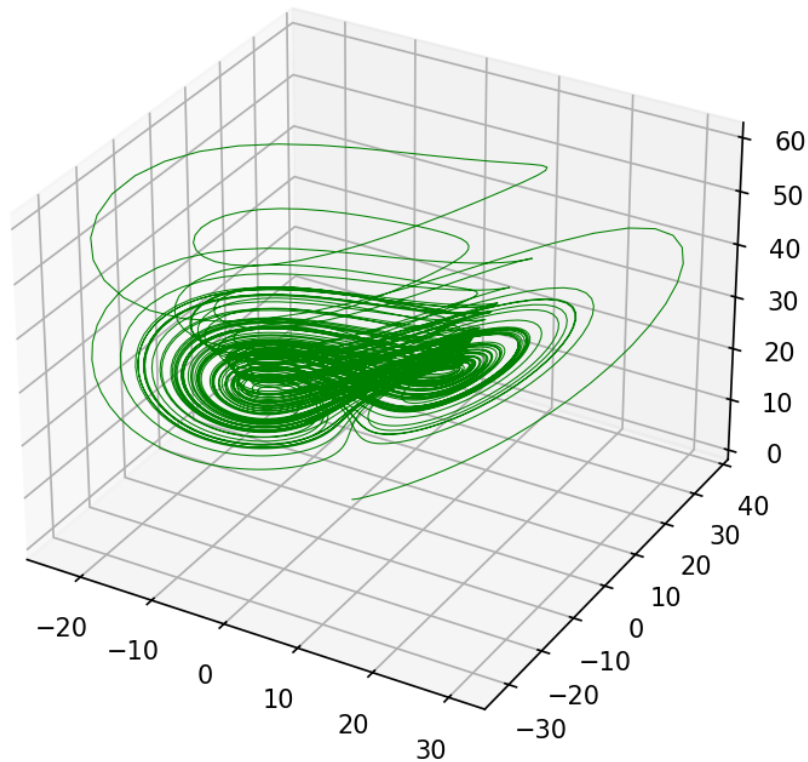
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3. Chen System

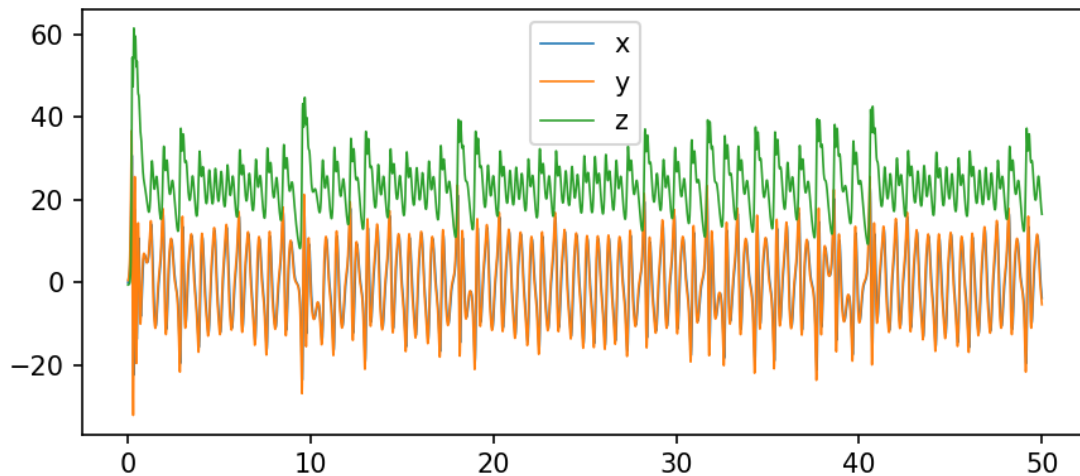
Equations: $\frac{dx}{dt} = a(y - x)$, $\frac{dy}{dt} = (c - a)x - xz + cy$, $\frac{dz}{dt} = xy - bz$

Parameters: $a=35$, $b=3$, $c=28$ | Initial Condition: $(-0.1, 0.5, -0.6)$ | $dt=0.005$ | Length: 50 | Solver: RK45

Chen Phase Space



Chen Time Series



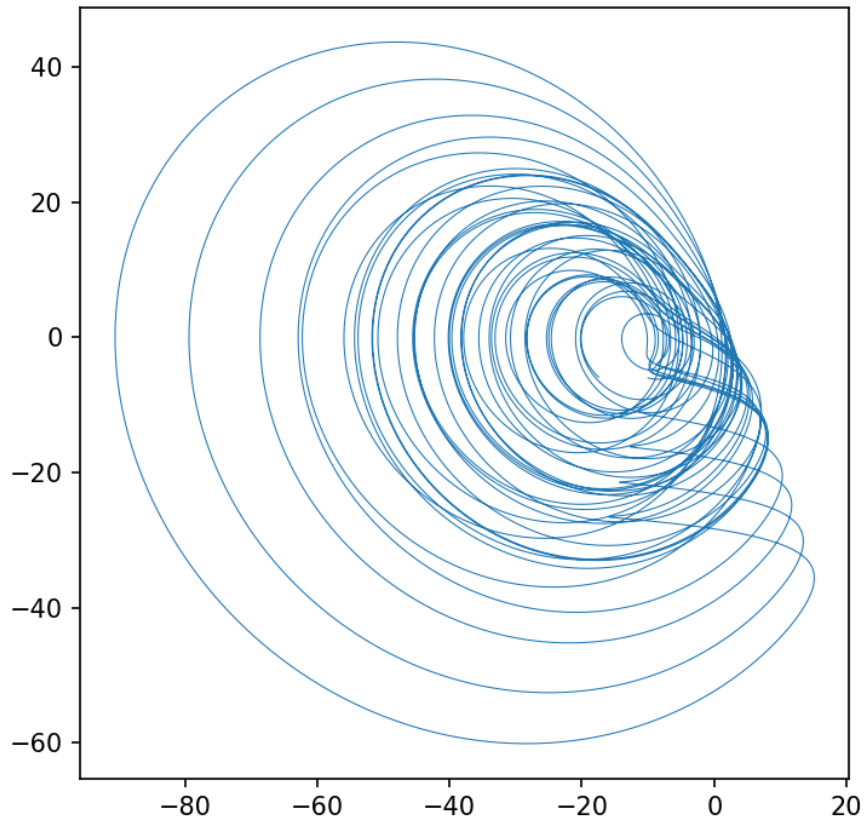
4. Hyperchaotic Rössler System

Equations: $\frac{dx}{dt} = -y - z$, $\frac{dy}{dt} = x + ay + w$, $\frac{dz}{dt} = b + xz$, $\frac{dw}{dt} = -cz + dw$

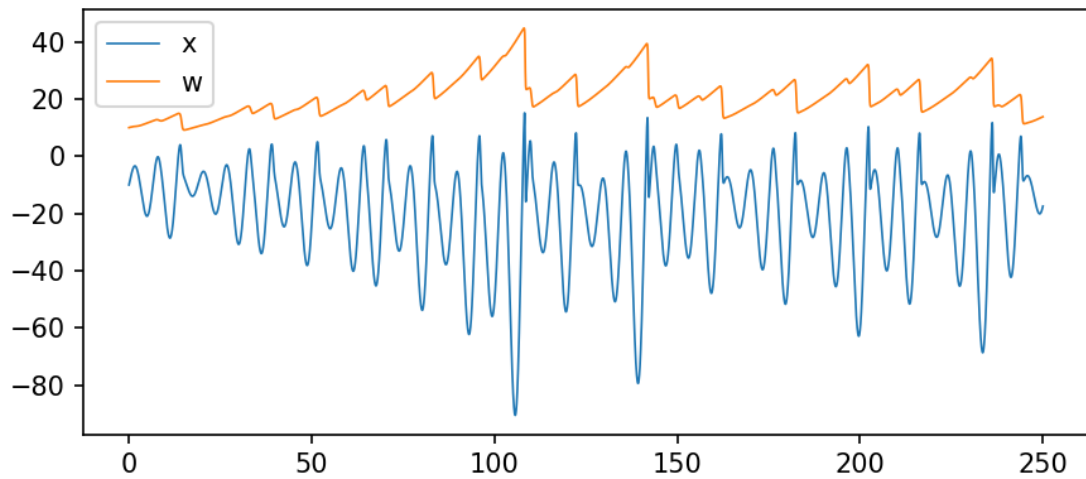
Parameters: $a=0.25$, $b=3$, $c=0.5$, $d=0.05$ | Initial Condition: $(-10, -6, 0, 10)$ | $dt=0.01$ | Length: 250 | Solver: RK45

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Hyperchaotic Rössler (x-y Projection)



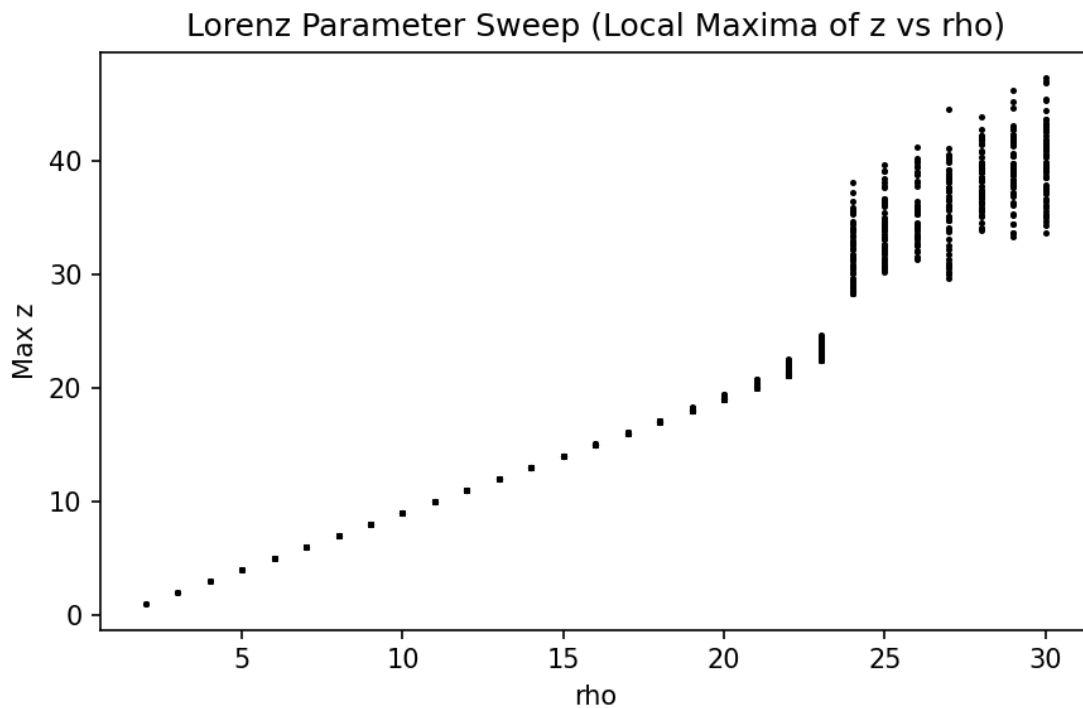
Hyperchaotic Rössler Time Series



Part C: Sensitivity Analysis

1. Parameter Sweep (Lorenz System)

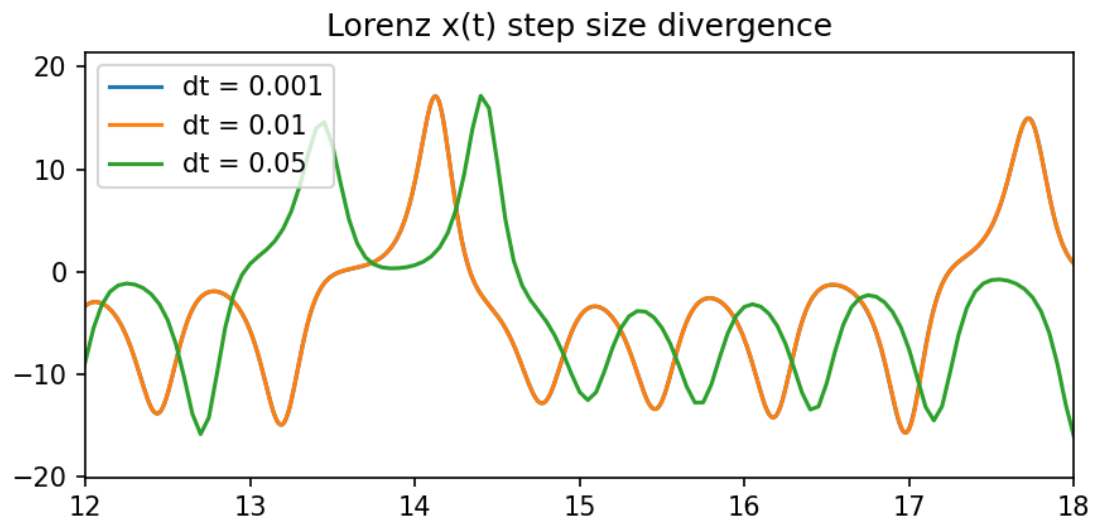
The parameter sweep diagram shows the local maxima of the z -variable as the parameter ρ increases. This is the continuous equivalent of a bifurcation diagram. It clearly illustrates the transition from a stable point to chaos, with small windows of periodic behavior visible as straight lines within the chaotic regions.



2. Step-size Sensitivity

This plot shows the Lorenz system simulated with three different time steps ($dt = 0.001, 0.01, 0.05$). Because chaotic systems are highly sensitive to initial conditions and numerical errors, the small differences caused by the step sizes accumulate over time. Eventually, the trajectories diverge completely, proving that smaller time steps (like $dt=0.001$ or 0.01) are necessary to accurately simulate chaotic systems.

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Conclusion

This report explored both discrete and continuous chaotic systems. Discrete systems like the Logistic and Henon maps demonstrate how simple algebraic rules can lead to complex, unpredictable behavior through period-doubling cascades. In contrast, continuous systems like Lorenz and Rössler use differential equations to create strange attractors in 3D phase space. Furthermore, the hyperchaotic Rössler system showed that adding a fourth dimension allows for even more complex dynamics, characterized by multiple positive Lyapunov exponents. Across all models, extreme sensitivity to parameters and initial conditions remains the fundamental signature of chaos.